

RW-5.0 Interface Control Document

Extra-Large Reaction Wheel - 5.0 Nms

Hardware Revision 2

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Product:	RW-5.0 Extra-Large Reaction Wheel
Hardware Revision:	2
Prepared By:	AnyCompany Engineering Team
Approved By:	Systems Engineering

1. Document History

Version	Date	Author	Description
1.0	2026-01-15	Systems Engineering	Initial release for first-flight qualification. Full mechanical, electrical, environmental, and software interface definition.
1.1	2026-03-22	Environmental Test	Updated shock specification to 1200G half-sine 0.3ms based on qualification test results.
1.3	2026-06-10	Avionics Team	Added dedicated health monitoring bus (J2 connector). Expanded message ID space to 16-bit. Added bearing health check and spectral analysis commands.

1.1 Approval Signatures

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1.2 Applicable Documents

ID	Title	Rev
AC-DS-00033	RW-5.0 Product Data Sheet	A
AC-TP-00033	RW-5.0 Qualification Test Procedure	B
NASA GSFC-STD-7000B	General Environmental Verification Standard (GEVS)	-
EIA/TIA-422-B	Balanced Voltage Digital Interface Circuits	-
EIA/TIA-485-A	Balanced Multipoint Systems	-
ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test	-
MIL-STD-810G	Environmental Engineering Considerations	-

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3. Scope and Applicability

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW-5.0 extra-large reaction wheel, hardware revision 2. It establishes the requirements and constraints that must be satisfied by both the reaction wheel assembly and the host spacecraft to ensure proper integration and operation.

The RW-5.0 is the largest single-axis reaction wheel in the AnyCompany product line, providing 5.0 Nms angular momentum storage with 150 mNm peak torque. It is designed for attitude determination and control subsystems (ADCS) on medium and large spacecraft. The design features dual-redundant motor windings (A/B selectable), a dedicated health monitoring processor, isolated power domains for motor, logic, and communications, and a bearing health monitoring system for predictive maintenance.

First flights are scheduled for 2026. The bearing design shares common heritage across all AnyCompany reaction wheel models, with 420+ cumulative flight years across 263 units.

3.1 Key Performance Summary

Parameter	Value	Units
Angular Momentum	+/- 5.0	Nms
Peak Torque	+/- 150	mNm
Mass	< 3000	g
Dimensions	194 x 192 x 57.25	mm
Supply Voltage (nominal)	28	V
Supply Voltage (range)	22 to 34	V
Nominal Power	20	W
Peak Power	120	W
Command Interface	RS-422 / Redundant RS-485	-
Baud Rate	921600	bps
Operating Temperature	-30 to +70	deg C
Vibration	14.1	Grms
Radiation (TID)	50	krad
Shock	1200	G (half-sine 0.3ms)
Control Mode	Speed, Torque	-
Heritage	First flights 2026; bearings: 420+ yrs, 263 units	-

4. Mechanical Interface

4.1 Mounting Interface

The RW-5.0 utilizes a bolt-circle mounting pattern for integration onto the spacecraft structure. The mounting face serves as the primary thermal interface; adequate thermal coupling must be maintained for proper heat rejection.

Parameter	Value
Mounting Pattern	12x M5 on 175mm bolt circle
Alignment Pins	3x dowel pins on mounting face
Lifting Points	4x M4 threaded inserts on top face
Connectors	J1 (51-pin) and J2 (15-pin) on housing side
Protective Covers	2x connector covers (remove before flight)
Thermal Interface	Mounting face is primary thermal sink
Torque Spec	M5 bolts: 3.5 Nm with Loctite 222

4.2 Mass and Dimensions

Cylindrical housing with a diameter of 194 mm and height of 57.25 mm. Total mass less than 3000 g.

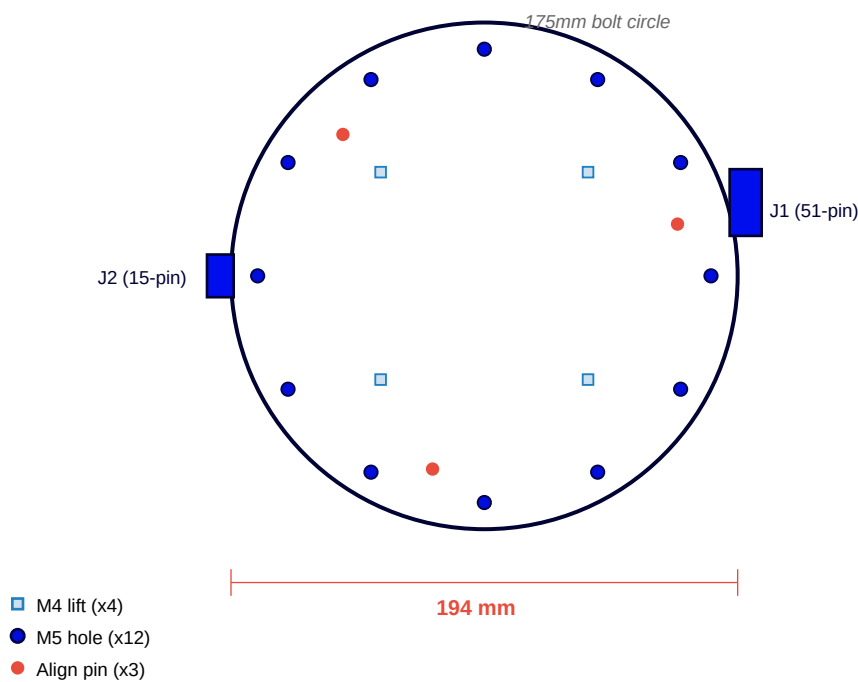


Figure 1: RW-5.0 Top View Mechanical Outline (not to scale)



Figure 2: RW-5.0 Product Photograph

4.3 Alignment and Handling

Three dowel pins enable precise angular clocking. Four M4 lifting points facilitate safe handling. Support at mounting face during transport; do not lift by housing wall.

4.4 Remove Before Flight Items

Item	Remove?	Notes
J1 connector cover	Yes	Red anodized, captive screws
J2 connector cover	Yes	Red anodized, captive screws
Bearing transport lock	Yes	4x screws on top face (marked RBF)
Desiccant bag	Yes	Inside shipping container

5. Environmental Specifications

5.1 Thermal

Parameter	Min	Typ	Max	Units	Notes
Survival Temp	-50	-	+95	deg C	Non-operating
Operating Temp	-30	+20	+70	deg C	At mounting interface
Cold Start Heater	-20	-	-	deg C	Integrated heater below -20C
Heater Power	-	-	15	W	3-zone, spacecraft controlled
Thermal Resistance	-	-	0.5	K/W	Mounting face

Table 1: Thermal Specifications

5.2 Pressure and Outgassing

Qualified for sea-level and hard vacuum. Not qualified for high-altitude atmosphere; do not power during ascent. TML < 1%, CVCM < 0.1% per ECSS-Q-ST-70-02C.

5.3 Vibration

Qualified to 14.1 Grms random vibration, 2 min/axis per NASA GEVS (GSFC-STD-7000B).

Frequency Range	Qual Level	Accept Level	Duration
20 - 50 Hz	+3 dB/oct	+3 dB/oct	2 min/axis
50 - 800 Hz	0.04 g ² /Hz	0.02 g ² /Hz	2 min/axis
800 - 2000 Hz	-6 dB/oct	-6 dB/oct	2 min/axis
Overall	14.1 Grms	10.0 Grms	-

Table 2: Random Vibration Specification

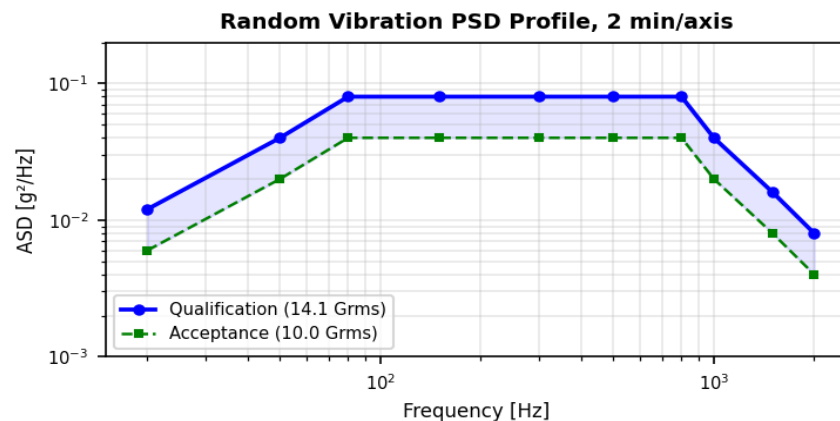


Figure 3: Random Vibration PSD Profile

5.4 Radiation

Radiation-hardened components. TID: 50 krad at board level. SEE mitigation: EDAC on critical memory, watchdog-protected processors.

5.5 Shock

Qualified to 1200G half-sine, 0.3ms pulse. Highest shock rating in AnyCompany line. 3 shocks/axis both directions per MIL-STD-810G.

6. Electrical Interface

6.1 Power Architecture

FOC 3-phase BLDC with dual-redundant windings (A/B selectable). Dedicated health processor on isolated domain. Active bus regulation with regenerative recovery. Bus clamp at 36V.

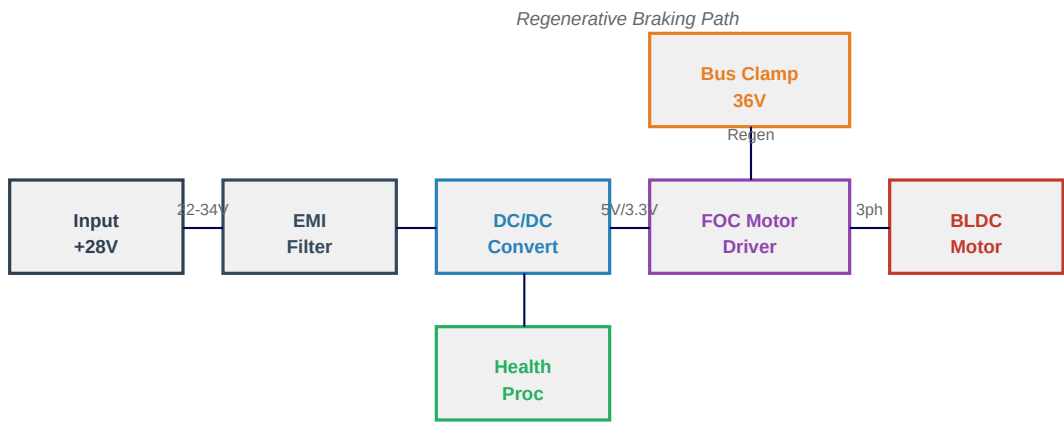


Figure 4: Power Architecture Block Diagram

6.2 Connector J1: Main Interface (51-pin Micro-D)

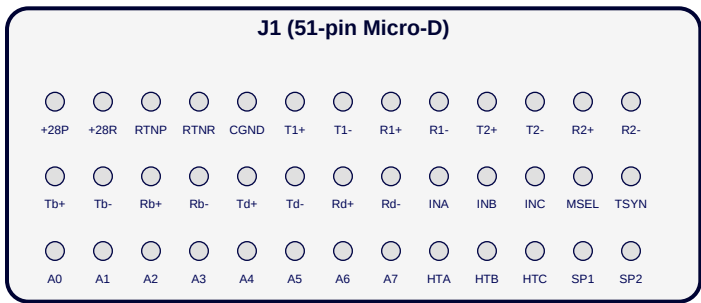


Figure 5: J1 Connector Face View

Pin(s)	Signal	Color	Description	Voltage
1-2	+28V PRI/RED	Red/Red	Primary/Redundant power	22-34V
3-4	RTN PRI/RED	Black/Black	Power returns	0V ref
5	Chassis GND	Green	Chassis/EMI ground	-
6-9	RS-422 CH1	Org/Brn/Blu/Wht	Primary TxA+/-/RxA+/-	Diff +/-5V
10-13	RS-422 CH2	Org/Brn/Blu/Wht	Secondary Cmd/Telem	Diff +/-5V
14-17	RS-485 RED x2	Vio/Gry/Pnk/Tan	Redundant channels	Diff +/-5V
18-20	INHIBIT A/B/C	Yellow x3	Active low inhibits (triple)	0-3.3V
21	MOTOR_SELECT	White	Winding A/B selection	0-3.3V
22	TELEM_SYNC	Cyan	Telemetry sync pulse	0-3.3V

Pin(s)	Signal	Color	Description	Voltage
23-30	ANALOG_OUT x8	Gray x8	Analog telemetry outputs	0-5V
31-33	HEATER A/B/C	Red/Wht x3	3-zone heater enable	0-28V sw
34-51	SPARE	-	Reserved	-

Table 3: J1 Connector Pinout

6.3 Connector J2: Health Monitor (15-pin Micro-D)

J2 provides a dedicated health monitoring bus, independent of J1. Enables continuous bearing vibration monitoring even if the primary bus is saturated.

Pin(s)	Signal	Color	Description	Voltage
1-2	Health Tx+/-	Org/Brn	RS-422 Health transmit	Diff +/-5V
3-4	Health Rx+/-	Blu/Wht	RS-422 Health receive	Diff +/-5V
5	+28V Feed	Red	Health processor power	22-34V
6	RTN Feed	Black	Health power return	-
7-10	BRG_SNS 1-4	Gray x4	Bearing accelerometer signals	0-3.3V
11-15	SPARE	-	Reserved	-

Table 4: J2 Connector Pinout

6.4 Power Consumption

Condition	Power	Current @ 28V	Notes
Standby (motor off)	3 W	0.11 A	Logic and health processor only
Idle @ 3500 RPM	20 W	0.71 A	No torque, bearing friction
Nominal operation	40 W	1.43 A	Typical ADCS duty cycle
Peak torque (150 mNm)	120 W	4.29 A	Short duration, thermal limited
Regen braking	-15 W	-0.54 A	Current sourced to bus
Heaters (3-zone max)	15 W	0.54 A	Spacecraft controlled

Table 5: Power Consumption

7. Software Interface

7.1 Protocol Frame Format

Binary-Extended protocol with 16-bit message IDs (expanded from 8-bit in smaller AnyCompany wheels).

Binary-Extended Frame (max 2060 bytes)



Figure 6: Protocol Frame Structure

SYNC=0xEB90 big-endian. LENGTH=bytes after LENGTH. MSG_ID=16-bit. SEQ=16-bit wraps 0xFFFF. CRC32 poly 0x04C11DB7, init 0xFFFFFFFF, complement. All multi-byte little-endian except SYNC.

7.2 Communication Channels

Channel	Connector	Interface	Baud	Purpose
CMD/TELEM	J1 (pins 6-17)	RS-422/RS-485	921600	Commands, telemetry, FW updates
HEALTH	J2 (pins 1-4)	RS-422	921600	Bearing vibration, thermal, power quality

Table 6: Communication Channels

7.3 Message Identifiers

Main bus: 0x0001-0x00FF. Health bus: 0x1000-0x1003.

ID	Name	Dir	Description	Len
0x0001	HEARTBEAT	RW->SC	Periodic status	4B
0x0002	SET_MODE	SC->RW	Set operating mode	2B
0x0003	SET_SPEED	SC->RW	Speed setpoint	4B
0x0004	SET_TORQUE	SC->RW	Torque command	4B
0x0005	SET_MOMENTUM	SC->RW	Momentum setpoint	4B
0x0006	RAMP_PROFILE	SC->RW	Speed ramp trajectory	Var
0x0010	TELEM_HK	RW->SC	Housekeeping (10 Hz)	72B
0x0011	TELEM_SCIENCE	RW->SC	Science (200 Hz)	Var
0x0012	TELEM_JITTER	RW->SC	Jitter data	Var
0x0020	CONFIG_READ	SC->RW	Read config param	4B
0x0021	CONFIG_WRITE	SC->RW	Write config param	8B
0x0022	CONFIG_STORE	SC->RW	Store to NVM	0B
0x0023	CONFIG_LOAD	SC->RW	Load from NVM	0B
0x0030	FW_UPD_START	SC->RW	Begin FW update	8B
0x0031	FW_UPD_DATA	SC->RW	FW data block	Var

ID	Name	Dir	Description	Len
0x0032	FW_UPD_VERIFY	SC->RW	Verify FW CRC	4B
0x0040	SELF_TEST	SC->RW	Built-in self test	2B
0x0041	CALIBRATE	SC->RW	Calibration sequence	2B
0x0042	DESAT	SC->RW	Desaturation assist	4B
0x0043	ZERO_MOMENTUM	SC->RW	Drive to zero speed	0B
0x0044	BRG_HEALTH	SC->RW	Bearing health check	0B
0x0050	SPECTRAL	RW->SC	FFT spectral output	512B
0x00F0	ERROR	RW->SC	Error report	Var
0x00FF	RESET	SC->RW	Software reset	0B
0x1000	HLTH_STATUS	RW->SC	Health status (J2)	16B
0x1001	BRG_VIB	RW->SC	Bearing vib spectrum (J2)	512B
0x1002	THERMAL_MAP	RW->SC	Thermal gradient (J2)	32B
0x1003	PWR_QUALITY	RW->SC	Power ripple (J2)	16B

Table 7: Message Identifiers

7.4 Operating Modes

Mode	ID	Description	Entry Condition
STANDBY	0x00	Motor off, telem active	Power-on default
SPEED_CTRL	0x01	Closed-loop speed	SET_MODE from STANDBY
TORQUE_CTRL	0x02	Open-loop torque cmd	SET_MODE from STANDBY
MOMENTUM_CTRL	0x03	Closed-loop momentum	SET_MODE from STANDBY
SAFE	0x10	Motor braked, minimal telem	Fault or SET_MODE
FW_UPDATE	0x20	Firmware update	FW_UPD_START from STANDBY
SELF_TEST	0x30	Built-in test	SELF_TEST from STANDBY
DESAT_ASSIST	0x40	Desaturation profile	DESAT from ctrl mode
BRG_RUNIN	0x50	Bearing conditioning	SET_MODE from STANDBY

Table 8: Operating Modes

8. Telemetry Definitions

8.1 Housekeeping Packet (0x0010, 10 Hz)

Offset	Field	Type	Units	Range
0	wheel_speed_rpm	float32	RPM	+/- 3500
4	torque_actual_mNm	float32	mNm	+/- 150
8	current_phaseA_mA	float32	mA	0-5000
12	current_phaseB_mA	float32	mA	0-5000
16	current_phaseC_mA	float32	mA	0-5000
20	bus_voltage_V	float32	V	0-40
24	bus_current_A	float32	A	-1 to +5
28	pcb_temp_C	float32	deg C	-50 to +100
32	motor_temp_A_C	float32	deg C	-50 to +100
36	motor_temp_B_C	float32	deg C	-50 to +100
40	bearing_temp_A_C	float32	deg C	-50 to +100
44	bearing_temp_B_C	float32	deg C	-50 to +100
48	fault_register	uint32	bitfield	See Sec 9
52	mode_state	uint8	-	See Table 8
53	uptime_sec	uint32	seconds	0 to 2 ³²
57	cumulative_revs	uint32	revs	0 to 2 ³²
61	power_mWh	uint32	mWh	0 to 2 ³²
65	momentum_Nms	float32	Nms	+/- 5.0
69	jitter_rms	float32	arcsec	0-100

Table 9: Housekeeping Telemetry Packet

8.2 Health Monitoring Packet (0x1001, 1 Hz on J2)

Offset	Field	Type	Units	Notes
0	bearing_vib_spectrum	uint16[256]	mg	256-bin FFT (0-5 kHz)
512	thermal_gradient	float32	deg C	Max temp difference
516	power_ripple_mV	float32	mV	Peak-to-peak ripple
520	insulation_Mohm	float32	Mohm	Winding insulation

Table 10: Health Monitoring Packet

9. Fault Register

32-bit register at HK offset 48. Bits latched until RESET (0x00FF) or power cycle.

Bit	Name	Description	Recovery
0	OC_PH_A	Overcurrent phase A	Auto-retry 3x, then SAFE
1	OC_PH_B	Overcurrent phase B	Auto-retry 3x, then SAFE
2	OC_PH_C	Overcurrent phase C	Auto-retry 3x, then SAFE
3	OV_BUS	Bus overvoltage (>36V)	Bus clamp; SAFE if persist
4	UV_BUS	Bus undervoltage (<20V)	SAFE mode
5	OT_PCB	PCB overtemp (>85C)	Torque derate; SAFE >95C
6	OT_MOT_A	Motor A overtemp (>90C)	Switch to winding B
7	OT_MOT_B	Motor B overtemp (>90C)	SAFE if both hot
8	OT_BRG_A	Bearing A overtemp (>80C)	Speed limit applied
9	OT_BRG_B	Bearing B overtemp (>80C)	SAFE mode
10	HALL	Hall sensor fault	Sensorless fallback
11	ENCODER	Encoder fault	Back-EMF speed estimate
12	COMM_PRI	Primary comm timeout >2s	Switch to redundant
13	COMM_RED	Redundant comm timeout >2s	SAFE mode
14	STALL	Motor stall detected	Auto-restart 3x
15	RUNAWAY	Speed >+/-4000 RPM	Emergency brake, SAFE
16	MEM	EDAC uncorrectable error	Reset and reload NVM
17	WDOG	Watchdog timer expired	Auto-reset, log event
18	BRG_DEG	Bearing degradation alarm	Operator notification
19	INSUL	Winding insulation low	Switch winding, notify
20	HLTH_BUS	Health bus (J2) lost	Main bus reports status
21	REDUN_SW	Redundancy switchover	Informational
22-31	EVT_CNT	10-bit event counter	Wraps at 1023

Table 11: Fault Register Bitfield

10. Register and Memory Map

Accessible via CONFIG_READ (0x0020) and CONFIG_WRITE (0x0021). R = read-only.

Address	Field	Type	Access	Description
0x0000	speed_setpoint	float32	RW	Speed command (RPM)
0x0004	torque_setpoint	float32	RW	Torque command (mNm)
0x0008	momentum_setpoint	float32	RW	Momentum command (Nms)
0x000C	mode_command	uint8	RW	Operating mode ID
0x0010	motor_select	uint8	RW	0=Winding A, 1=Winding B
0x0014	speed_kp	float32	RW	Speed loop proportional gain
0x0018	speed_ki	float32	RW	Speed loop integral gain
0x001C	torque_limit	float32	RW	Max torque limit (mNm)
0x0020	speed_limit	float32	RW	Max speed limit (RPM)
0x0024	watchdog_timeout	uint16	RW	Watchdog timeout (ms)
0x0028	telem_rate_hk	uint16	RW	HK telem rate (Hz)
0x002A	telem_rate_sci	uint16	RW	Science telem rate (Hz)
0x0030	brg_alarm_thresh	float32	RW	Bearing vib alarm (mg)
0x0034	thermal_lim_pcb	float32	RW	PCB temp limit (deg C)
0x0038	thermal_lim_motor	float32	RW	Motor temp limit (deg C)
0x003C	thermal_lim_brg	float32	RW	Bearing temp limit (deg C)
0x0100	serial_number	uint32	R	Factory serial number
0x0104	fw_version	uint32	R	Firmware version
0x0108	hw_revision	uint8	R	Hardware revision
0x010C	cal_date	uint32	R	Last calibration (UNIX time)
0x0110	operating_hours	uint32	R	Cumulative hours
0x0114	total_revolutions	uint64	R	Lifetime rev count

Table 12: Register and Memory Map

11. Bearing Health Monitoring

The RW-5.0 includes a dedicated bearing health monitoring subsystem for predictive maintenance and early detection of bearing degradation.

11.1 System Overview

The health processor samples four bearing accelerometers at 10 kHz. A 1024-point FFT is computed every 100 ms, producing a 256-bin magnitude spectrum (0 to 5 kHz). The spectrum is transmitted via J2 as message 0x1001 at 1 Hz. Raw accelerometer data is available on J2 BRG_SNS pins for spacecraft-level analysis.

11.2 Monitored Frequencies

Characteristic bearing defect frequencies are monitored. Alarm thresholds configurable via CONFIG_WRITE (register 0x0030). Values shown for 3500 RPM.

Frequency	Abbrev.	Formula	Value @ 3500 RPM	Alarm
Cage	FTF	$0.4 \times \text{RPM} / 60$	23.3 Hz	5 mg
Ball pass inner	BPFI	$N_b \times 0.6 \times \text{RPM}/60$	122.5 Hz	8 mg
Ball pass outer	BPFO	$N_b \times 0.4 \times \text{RPM}/60$	81.7 Hz	8 mg
Ball spin	BSF	$(D_p/d_b) \times \text{RPM}/60$	35.0 Hz	5 mg
2x Ball spin	2xBSF	$2 \times \text{BSF}$	70.0 Hz	4 mg

Table 13: Bearing Characteristic Frequencies

11.3 Alarm Logic

When any monitored bin exceeds its threshold for 10 consecutive spectra (1 second), BRG_DEGRADE fault bit (bit 18) is set. Reported on both health bus (0x1000) and main bus HK. No autonomous mode change; operator decides response.

11.4 Trending and Ground Analysis

The health processor maintains a 24-hour rolling average of each monitored frequency magnitude. Trend data available via CONFIG_READ at registers 0x0200-0x0220. Ground analysis enables detection of gradual wear patterns before alarm thresholds. Recommend downlinking health packets at least once per orbit for long-duration missions.

End of Document